Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			aluminium ions gain electrons therefore are reduced (1) accept $Al^{3+} + 3e^- \rightarrow Al$ ⇨ reduction oxide ions lose electrons therefore are oxidised (1) accept $2O^{2-} \rightarrow O_2 + 4e^-$ ⇨ oxidation neutral answer – oxidation is loss, reduction is gain	2			2		
	(b)	(i)		$\frac{36}{100} \times 500 = 180$ tonnes of Al_2O_3		1		1	1	
		(ii)		95.3 / 95 (3) if answer incorrect credit each correct step in method 102 tonnes Al_2O_3 produces 54 tonnes of Al (1) 1 tonne Al_2O_3 produces $\frac{54}{102}$ tonnes of Al (1) 180 tonnes Al_2O_3 produces 95.3 tonnes of Al (1) ecf possible from part (i)		3		3	3	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				alternative method $\frac{180}{102}$ = 1.765 (1) 1.765 × 2 = 3.530 (1) 3.530 × 27 = 95.3 (1) ecf possible from part (i)						
				Question 9 total	**2**	**4**	**0**	**6**	**4**	**0**